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(1) Research Program/Project/Study Title:		Acceptability of Jackfruit (<i>Artocarpus heterophyllus</i>) Siomai
Proponent Name:		Meldred D. Lagon
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Program or Project has component (if no proceed to 2)? ☐ Yes / No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences / Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center /FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Sigma
Province:	Capiz
Region:	VI
(5) Classification Research: Development: / Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation / Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Agriculture	(8) Discipline/Field of Specialization Food

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Background of the Study

Swapping meat for plants reduces saturated fat and increases the fiber and vitamin content of dishes. Studies show that this leads to reduced risks for diabetes, cancer, and heart disease. Plants require fewer environmental resources like water and space, which can offset climate change.

Jackfruit is a unique fruit that is commonly used as a meat alternative. Its stringy texture is said to resemble pork and it is often used as such by vegans and vegetarians. Eating less meat is not only better for one's health but also for the environment. It is considered as the largest edible fruit in the world and found to contain large quantities of nutrients, such as carbohydrates, proteins, vitamins, minerals, and phytochemicals. Studies by various research groups documented the anti-carcinogenic, antimicrobial, antifungal, anti-inflammatory, wound healing, and hypoglycemic effects along with digestive and immunological benefits of jackfruit (Sowmyashree and Devaraja, 2022).

A growing need for delicious, nutritious and convenient food encourages the demand for siomai. The classic siomai is never outdated because its filling can be any ingredients to satisfy difference dietary needs. It is a traditional Chinese dumpling and has made its way to the heart of the Filipino's as evidenced by the hundreds of stalls, eateries, and restaurants who serve them. It Basically makes use of ground pork, shrimps, and mushrooms as a main ingredients (Merano, 2023).

Vegetables play an important role in human nutrition. Most are low in fat and calories but are bulky and filling. They supply dietary fiber and are important sources of essential vitamins, minerals and trace elements. Particularly important are the antioxidant vitamins A, C and E. When vegetables are included in the diet, there is found to be a reduction in the incidence of cancer, stroke, cardiovascular disease and other chronic ailments. With this above-mentioned claims, the researcher wanted to utilize indigenous vegetable found in the locality like jackfruit in making siomai.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The proposed study on acceptability of Jackfruit (*Artocarpus heterophyllus*) siomai will utilized young unripe jackfruit as a meat substitute in making siomai. The young unripe jackfruit will be peel, clean, chop into small pieces, boil, and drain. It will then use as substitute to ground pork in making siomai using five treatments with varying proportions.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives of the Study

This study will be experimental in nature and its purpose is to determine the general acceptability of jackfruit siomai. Specifically, it has the following objectives:

1. to determine the sensory qualities of the product in terms of appearance, aroma, taste, and texture;
2. to determine which treatment would produce a generally acceptable siomai using the following proportions:
 - Treatment A- 100 % ground pork
 - Treatment B- 75% ground pork and 25 % ground jackfruit
 - Treatment C- 50% ground pork and 50 % ground jackfruit
 - Treatment D- 25% ground pork and 75 % ground jackfruit
 - Treatment E- 100 % ground jackfruit and;
3. to determine if there are significant differences in the sensory qualities and general acceptability of the product among Treatments.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of – the-art of current technology / information from which the project proposal will take-off).

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Review of Related Literature/Prior Art Search

Result of the study of Alayon et al. (2018) their utility model disclosed wherein bamboo shoots is substituted for meat in making siomai. The ingredients of the siomai comprise flour, eggs, onions, garlic, turnips, carrots, pork, and bamboo shoots. The procedure of cooking the siomai comprises washing the bamboo shoots, boiling the bamboo shoots, straining, mincing the bamboo shoots, mixing the chopped ingredients in a bowl, placing the mixture into the wrapper, folding the wrapper to create a round dumpling, and finally, steaming.

Ariza et al. (2020) with Utility Model PH22020000142U1 Process of Making Mackerel (*Rastrelliger Brachysoma*) Siomai In this utility model, a mackerel (*Rastrelliger brachysoma*) siomai is disclosed. The siomai is composed of 4.96 cups mackerel flakes, 1 pc medium egg, 1 pack molo wrapper, 0.47 cup salt, 0.47 cup chilled water, 0.31 cup flour, 0.31 cup ground black pepper, 0.125 chopped garlic, _ cup carrots, √ cup chopped young green onion, « cup grated cheese, 0.21 cup paprika and 0.125 cup sesame oil.

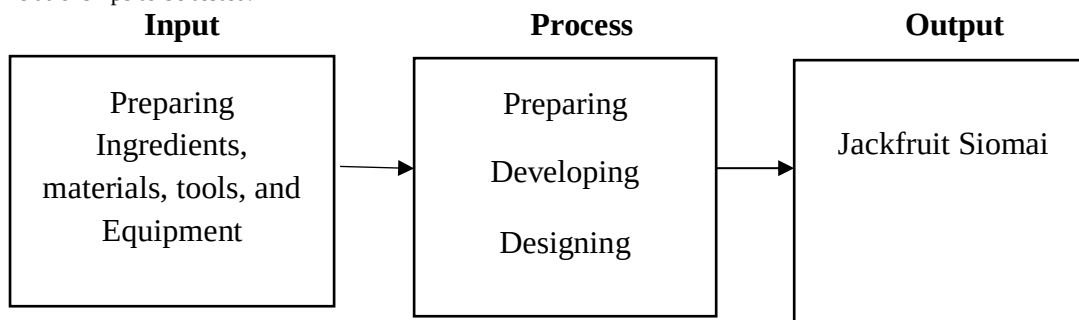
Siomai is one of the favorite snacks or a dish of Filipino nowadays. It is a traditional Chinese dumpling which is adopted and embraced. This popular dumpling has made its way on the heart of the Filipino's as proof by the hundreds of stalls and eateries who serve them. A dumpling (siomai) has a filling of pork, beef or a shrimp along with other vegetables and sauce; it is a dish that consists of small pieces of dough that made from variety of starch, often wrapped with filling. As the popularity increase of this dish, there are people who create their own recipe just to be different from the other (Domingo et al., 2017).

The origin of siomai could be traced back to the Yuan Dynasty around 700 yrs ago. It was invented in the northern regions of China and then slowly spread it spread down south. It is a traditional Chinese dumpling and has made its way to the heart of the Filipino's as evidenced by the hundreds of stalls, eateries, and restaurants who serve them. It Basically makes use of ground pork, shrimps, and mushrooms as a main ingredients (Merano, 2023).

Artocarpus heterophyllus which is commonly known as jackfruit is a tropical climacteric fruit, belonging to Moraceae family, is native to Western Ghats of India and common in Asia, Africa, and some regions in South America. It is known to be the largest edible fruit in the world. Jackfruit is rich in nutrients including carbohydrates, proteins, vitamins, minerals, and phytochemicals. Both the seeds and the flesh of jackfruit are consumed as curries and boiled forms, while the flesh in fully ripen stage can be eaten directly as a fruit. Several countries have developed different food products such as jam, jellies, marmalades, and ice creams using pureed jackfruit. The several parts of jack tree including fruits, leaves, and barks have been extensively used in traditional medicine due to its anti-carcinogenic, antimicrobial, antifungal, anti-inflammatory, wound healing, and hypoglycemic effects. Despite all these benefits, unfortunately, the fruit is underutilized in commercial scale processing in regions where it is grown. The aim of this review is to disseminate the knowledge on nutritional and health benefits of jackfruit, in order to promote utilization of jackfruit for commercial scale food production Ranasinghe et al. (2019).

Jackfruit is a tropical fruit native to India. It commonly grows in the naturally warm and humid climates of Asia, Africa and some parts of South America. It's part of the same family as figs, mulberries and breadfruit. Jackfruit is a large tropical fruit that has a slightly sweet and fruity flavor and a meaty texture. It's often used as a replacement for meat in vegetarian or vegan dishes. One cup (150 grams) of jackfruit pieces contains the following: Calories: 143, Fat: 1 gram, Protein: 3 grams, Carbs: 35 grams, Fiber: 2 grams, Vitamin B6: 29%, Vitamin C: 23% (Feller and Petre, 2022).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.



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(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Methodology

This study will utilize the experimental research design to determine the sensory qualities such as appearance, aroma, taste, and texture; and general acceptability of utilizing young unripe jackfruit in making siomai.

The experimental design to be use in the study will be the single factor experiment with Completely Randomized Design (CRD) in which unripe jackfruit will be utilize in making siomai. In this study, the unripe ground jackfruit and ground pork will be added in different proportions; Treatment A – 100% ground pork, Treatment B – 75% ground pork and 25% ground jackfruit, Treatment C – 50% ground pork and 50% ground jackfruit, Treatment D – 25% ground pork and 75% ground jackfruit, and Treatment E –100% ground jackfruit.

Procedure in preparing young unripe jackfruit includes: peeling, cleaning, slicing, chopping, grounding, boiling, and draining. Procedure in making Jackfruit siomai are the following: First, combine ground pork, ground jackfruit, sliced carrots, chopped onions, soy sauce, egg, garlic, salt, pepper, and cornstarch. Then, mix until all ingredients combine. Scoop 1 Tablespoon of mixture and place in wonton wrapper. Steam for about 15 – 20 minutes. Cool, serve or package.

The arithmetic mean and Analysis of Variance (ANOVA) will be the statistical tools to be used in analyzing and interpreting the data. Using SPSS for data processing and analysis. The mean will be used to determine the appearance, aroma, taste, and texture of the product. The ANOVA set at 0.05 alpha level will used to determine the significant differences on the sensory qualities and general acceptability of the product among treatments.

The study will used a scorecard using a Nine-point Hedonic Scale. It looked into the four sensory qualities such as appearance, aroma, taste, and texture as the variables. These four sensory qualities will determine the general acceptability of jackfruit siomai which will be the dependent variable. The evaluators will be 30 semi-skilled panelist composing faculty teaching food-related subjects and Food Service Management students.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use Gantt chart.

Activities	February 2024	March 2024	April 2024	May 2024	June-July 2024	August 2024	September 2024
Proposal/ Presentation							
Preparation of manuscript and Instrument							
Gathering of Data							
Data Encoding							
Interpretation of Data							
Presentation of Completed Research							

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Jackfruit Siomai

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(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Jackfruit Siomai, IEC materials

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

Children, kids, vege siomai lover, vegetarian people, and health conscious individual

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		1,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		1,000.00
- Meal allowance during data gathering		1,000.00
- Professional fees (laboratory analysis and other related fees)		9,000.00
- Miscellaneous (for incidental expenses)		1,000.00
- Printing		2,000.00
C. CAPITAL OUTLAY		
Total		15,000.00

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Alayon et al. (2018). PH22018050508U1 Composition of Bamboo Shoot Siomai Capiz State Univ Mambusao Satellite College [PH].

Feller, M. and Petre, A. (2022). Jackfruit: What Is This Vegetarian Meat Alternative? <https://www.healthline.com/nutrition/jackfruit-meat-alternative>

Ranasinghe, R., Maduwanthi, S., & Marapana, R. (2019). Nutritional and Health Benefits of Jackfruit (Artocarpus heterophyllus Lam.): A Review. International Journal of Food Science. <https://doi.org/10.1155/2019/4327183>

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<https://bigideaventures.com/plant-based-meat-alternatives-karanas-jackfruit-substitute-is-a-hit-in-dumplings-rillette-and-even-ngoh-hiang/>

Ariza et al. (2020). PH22020000142U1 Process of Making Mackerel (Rastrelliger Brachysoma) Siomai Eastern Visayas State Univ [PH]

Lawler, M. (2023). All about Jackfruit: Nutrition, Benefits, Weight Loss effect, Recipes, and more. <https://www.everydayhealth.com/diet-nutrition/jackfruit-nutrition-benefits-weight-loss-effect-recipes-and-more/>

<https://www.philstar.com/headlines/climate-and-environment/2022/08/28/2205556/filipino-appetite-sustainable-alt-meat-growing-bigger-push-needed>

(21) Capsule Curriculum Vitae (One-page brief C.V. only)